

MORDEN, MB, CanadaWMO: **715640**Lat: **49.1875N**Lon: **98.0839W**Elev: **298**StdP: **97.80**Time Zone: **-6.00 (NAC)**Period: **00-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-28.1	-25.8	-32.2	0.2	-27.9	-29.8	0.2	-25.5	10.1	-5.0	9.2	-7.5	3.1	280	0.615

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.7	30.6	21.5	29.0	20.3	27.4	19.2	23.4	28.2	22.0	26.9	20.7	25.4	4.4	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.9	17.2	26.2	20.3	15.5	24.9	19.0	14.3	23.7	71.5	28.5	66.0	27.0	61.2	25.5	29.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.0	8.1	7.2	-31.9	34.4	2.6	1.7	-33.8	35.6	-35.3	36.6	-36.7	37.5	-38.6	38.8		
(5)			-32.1	25.4	2.5	2.1	-33.9	26.9	-35.4	28.2	-36.8	29.3	-38.6	30.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	4.3	-12.9	-12.6	-4.3	4.7	11.7	17.5	20.4	19.3	14.3	6.2	-2.5	-11.0
	DBStd	13.40	8.17	7.15	7.43	5.82	5.01	3.57	3.12	3.43	4.55	5.12	6.61	7.28
	HDD10.0	3183	711	632	444	175	41	1	0	0	12	142	376	649
	HDD18.3	5303	969	865	702	409	212	57	16	29	134	377	625	908
	CDD10.0	1112	0	0	1	16	93	224	322	288	142	24	1	0
	CDD18.3	190	0	0	0	0	7	31	80	58	14	1	0	0
	CDH23.3	1802	0	0	0	3	103	290	681	552	159	13	0	0
	CDH26.7	493	0	0	0	0	26	62	185	177	41	3	0	0
(14) Wind	WSAvg	3.6	3.7	3.6	3.7	3.9	3.9	3.4	3.1	3.1	3.4	3.7	3.7	3.6
(15) Precipitation	PrecAvg	542	20	19	32	47	67	81	72	72	53	36	24	24
	PrecMax	784	65	63	110	141	220	189	151	198	158	123	57	48
	PrecMin	291	0	2	6	0	2	14	8	11	6	2	3	3
	PrecStd	121	14	16	26	39	52	41	40	49	36	31	16	13
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.7	5.7	16.4	23.4	30.0	30.6	32.5	33.4	30.4	25.1	15.8	6.1
		MCWB	3.1	2.5	9.8	12.8	16.7	22.0	23.2	22.5	19.9	15.4	10.1	2.9
	2%	DB	3.4	3.0	11.8	20.0	26.1	28.4	30.3	30.7	27.4	20.4	11.6	3.2
		MCWB	1.3	1.1	7.6	11.4	14.5	18.8	22.4	21.0	18.3	12.8	6.8	1.0
	5%	DB	1.2	0.5	7.8	17.4	23.4	26.5	28.8	28.5	24.9	17.0	8.8	0.9
		MCWB	-0.1	-0.7	4.7	9.7	13.4	18.2	21.5	20.2	17.0	11.0	5.3	-0.5
	10%	DB	-1.2	-1.7	4.8	14.5	20.6	24.7	27.2	26.5	22.2	14.3	6.2	-0.9
		MCWB	-2.2	-2.7	2.6	8.5	12.8	17.4	20.1	18.8	15.7	9.7	3.6	-1.8
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.2	2.7	11.9	14.9	18.7	24.2	26.0	24.7	22.2	16.3	10.4	3.2
		MCDB	5.5	5.0	14.9	19.8	24.9	28.5	30.1	29.8	28.1	22.4	14.9	5.7
	2%	WB	1.4	1.2	7.9	12.2	16.8	21.7	23.9	22.6	19.8	13.8	7.5	1.1
		MCDB	3.1	2.8	12.1	18.4	21.5	26.0	28.2	28.2	24.1	18.9	10.9	2.7
	5%	WB	-0.1	-0.5	4.8	10.4	15.2	19.6	22.6	21.2	18.2	11.7	5.5	-0.3
		MCDB	1.2	0.6	7.5	16.1	20.2	24.2	27.0	26.4	22.8	16.1	8.4	0.9
	10%	WB	-2.2	-2.7	2.8	8.8	13.8	18.2	21.2	19.9	16.7	9.9	3.7	-1.8
		MCDB	-1.2	-1.6	4.6	13.9	19.3	22.9	25.9	24.6	21.3	14.0	6.0	-0.9
(35) Mean Daily Temperature Range	MDBR		8.4	8.7	8.7	10.8	11.6	10.5	10.7	11.3	11.2	9.2	8.0	7.7
	5% DB	MCDBR	10.2	10.0	11.9	15.7	16.0	13.4	12.7	14.4	15.2	14.3	11.7	9.9
		MCWBR	8.8	8.4	8.4	8.9	7.7	6.6	6.8	6.8	7.7	7.9	8.0	8.5
	5% WB	MCDBR	9.9	9.7	11.4	14.9	12.2	11.5	11.9	12.6	13.2	13.3	11.1	9.4
		MCWBR	8.6	8.3	8.4	9.0	7.3	6.9	7.3	7.0	7.7	7.8	7.8	8.1
(40) Clear-Sky Solar Irradiance	taub		0.275	0.292	0.301	0.359	0.375	0.381	0.387	0.384	0.349	0.303	0.271	0.253
	taud		2.258	2.230	2.386	2.322	2.340	2.352	2.344	2.343	2.432	2.535	2.483	2.388
	Ebn at Noon		776	851	916	887	885	879	867	853	851	845	800	772
	Edh at Noon		72	95	97	117	120	119	118	113	93	71	58	56
(44) All-Sky Solar Radiation	RadAvg		1.49	2.69	3.93	4.97	5.53	5.95	6.32	5.20	3.80	2.32	1.32	1.05
	RadStd		0.13	0.21	0.33	0.38	0.41	0.42	0.27	0.34	0.27	0.31	0.18	0.12

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	-1.86	-2.30	N/A	N/A	N/A	N/A
Regional (28 neighbors)	N/A	N/A	N/A	N/A	N/A	-0.67	N/A	N/A	N/A	N/A